

NAME: _____ UIN: _____

QUIZ 2: (100 pts) (Close book/closed notes) Show all your work for full credit

1.(85 points) The power output of steam turbine is 5MW, and the inlet and the exit conditions are as follow: $P_1=2\text{MPa}$, $T_1=400^\circ\text{C}$, $V_1=50\text{m/s}$, $z_1=10\text{m}$; $P_2=15\text{kPa}$, $x_2=0.90$, $V_2=180\text{m/s}$, $z_2=6\text{m}$. a) Compare the magnitudes of Δh , ΔKE , ΔPE . b) Determine the work done per unit mass of the steam flowing through the turbine. c) Calculate the mass flow rate of the steam.

2. (15 points) Briefly explain the process that Lanzatech patented. (Monday's guest speaker; Rachel Brenc)

$$\left| \frac{dE_{cv}}{dt} = \dot{Q}_{cv} - \dot{W}_{cv} + \sum_i \dot{m}_i \left(h_i + \frac{V_i^2}{2} + gz_i \right) - \sum_e \dot{m}_e \left(h_e + \frac{V_e^2}{2} + gz_e \right) \right. \quad \left. \dot{m} = \frac{AV}{v} \right.$$

1 joule= 1 N.m

1 N= 1 kg.m/s²